

High- compared with low-dose radiosurgery for uveal melanomas

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Object. The authors compared the results of gamma knife radiosurgery in patients with uveal melanoma who underwent high-dose (treated from 1992–1995) and low-dose irradiation (treated from 1996–2002).

Methods. Thirty-one patients with uveal melanomas were treated with a mean margin dose of 52.1 Gy (high dose) and 33 with a mean dose of 41.5 Gy (low dose), and results were compared between groups. The technical procedure was the same in each group except for radiation dose.

In the low-dose group, complete tumor regression (scar formation) occurred in 12% and in the high-dose group in 26%. Partial regression (reduction of the tumor prominence between 50 and 80%) occurred in 81% of the low-dose group and in 58% of the high-dose group.

Neovascular glaucoma as a severe complication developed in 9% of the low-dose group and in 48% of the high-dose group.

Conclusions. Reduction of the margin dose from 52.1 to 41.5 Gy appears to achieve the same rate of tumor regression but is associated with a lower rate of severe side effects such as neovascular glaucoma. The follow-up period in the low-dose group, however, was much shorter.

KEY WORDS • gamma knife • radiosurgery • uveal melanoma

BETWEEN 1992 and 1995, 31 patients with uveal melanomas underwent high-dose GKS (~ 50 Gy to the margin).⁴ We found a high incidence of severe side effects of which neovascular glaucoma was the most serious. According to dose–volume histogram analysis performed during our treatments, significant factors for the development of complications were the tumor volume, location of the tumor, and a high radiation dose. In his series of GKS-treated vestibular schwannomas, Norén⁸ found an increase in the tumor control rate and a decrease of the rate of side effects after lowering the dose. Although we were treating cases of primary malignant tumor, dose reduction in some of our patients in whom a close follow up can be achieved seemed justified. In this study we have attempted to analyze the effect of a reduced dose on the rates of tumor regression and severe side effects.

Clinical Material and Methods

Technical Procedure

The GKS procedure performed at the Gamma Knife Center in Graz has been detailed previously.⁴ In our previous study we found that risk factors for the development

of neovascular glaucoma included tumor proximity to the ciliary body and tumor volume. A juxtapapillary location was the main risk factor for a radiation-induced optic neuropathy. Consequently we excluded cases involving tumors larger than 10 mm prominence and 16 mm base from single-fraction GKS. Moreover, juxtapapillary and ciliary body melanomas came to be considered as relative contraindications. In the following years we have lowered the dose to 30 Gy. Because both the high dose and the location at the ciliary body were risk factors, it was assumed that the chance of producing a neovascular glaucoma would be less if the dose were reduced. We treated one ciliary body melanoma with a dose of 35 Gy. Neovascular glaucoma has not developed in a follow-up period of 23 months. Longer follow-up study will show whether ciliary body melanomas can be reconsidered for GKS in the future (Figs. 1 and 2).

Patient Population

In the high-dose group (31 patients treated between 1992 and 1995) a mean margin dose of 51.1 Gy was given with a maximum dose to the center of the tumor of 120 Gy in one patient. The detailed results were previously reported.⁴ Seven patients were treated with a dose of 50 Gy or more, with one patient receiving 80 Gy to the tumor margin. In the low-dose group a margin dose of 45 Gy was

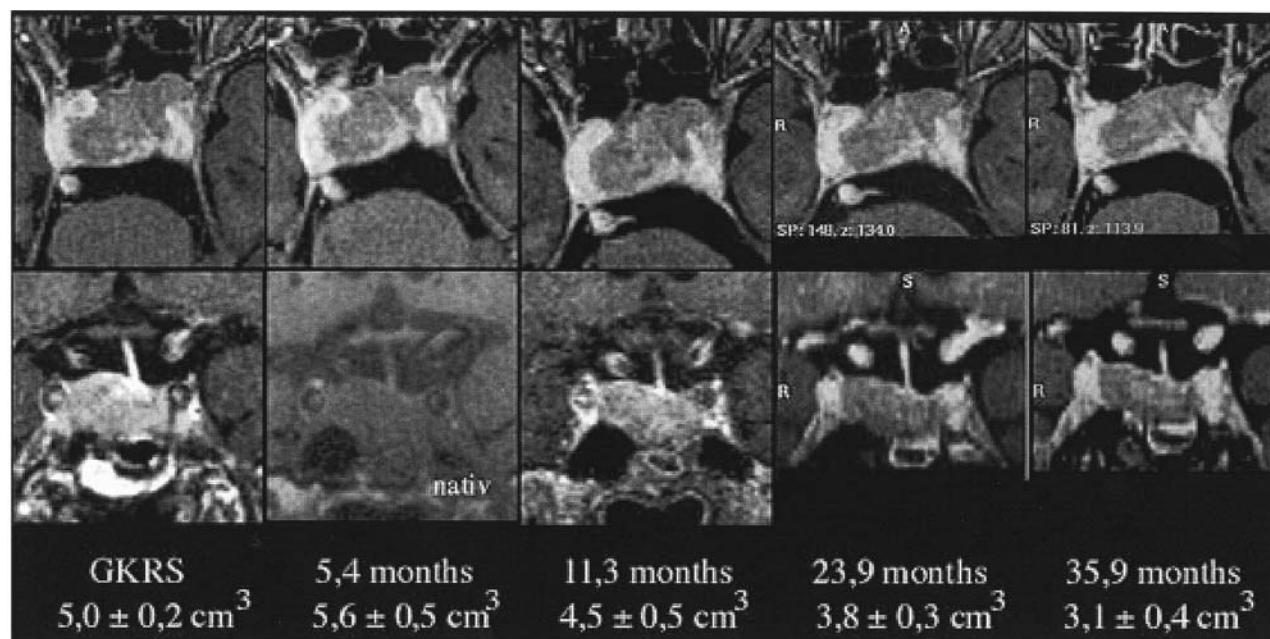


FIG. 1. Images obtained in a 35-year-old patient with amacular melanoma. Upper: Visual acuity prior to therapy was 20/25. Lower: Image obtained 24 months after single-fraction GKS. The margin dose was 35 Gy. Moderate side effects (radiation maculopathy) occurred and were treated by laser photocoagulation. Visual acuity 20/100.

given in 26 patients, nine patients received 40 Gy, 11 patients 35 Gy, and four patients received 30 Gy to the tumor margin. Most tumors were treated with the 8-mm collimators (thus facilitating the construction of a tumor-free margin of healthy tissue between 1–2 mm). The prescription dose was delivered to the 50 or 60% isodose in most patients (Table 1).

Results

The longest follow-up period was 79 months. The shortest was 12 months. The results are summarized in Table 2. In the high-dose group tumor regression (a decrease of the tumor prominence) occurred in 84%, and in the low-dose group in 93%. Total regression (scar formation) was distinguished from a partial regression, where a residual tumor prominence was observed. An attempt was made to determine the percentage of reduction of the tumor prominence. In the low-dose group, complete tumor regression (scar formation) occurred in 12% and in the high-dose group in 26%. Partial regression (50–80% reduction of the tumor prominence) occurred in 81% of the low-dose group and in 58% of the high-dose group.

Neovascular glaucoma as a severe complication occurred in 9% of the low-dose group and in 48% of the high-dose group.

The incidence of globe preservation in the high-dose group was 71% and in the low-dose group 72%. Recurrence occurred in five of 31 patients in the high-dose group and in five of 33 in the low-dose group. Proven metastatic disease developed in six patients in the high-dose group and two patients in the low-dose group. It was hard to determine the significance of the development of metastases in both groups because the follow-up period in the low-dose group has been much shorter.

Discussion

Achievements in Globe-Preserving Treatment

In 2002 preliminary results of the Collaborative Ocular Melanoma Study were reported.⁶ The authors attempted to determine whether globe-preserving treatment is justified or whether enucleation can achieve a better outcome in patients with uveal melanomas. In this, the largest multicenter trial ever undertaken, investigators could not show a benefit for enucleation in terms of the patient survival.⁶ Thus, it may be argued that enucleation is a treatment of the last resort.

Transpupillary Thermotherapy

In 1995 Oosterhuis, et al.,⁹ introduced a new treatment

TABLE 1
Patient characteristics and treatment parameters

Parameter	Mean ± Standard Deviation (%)	Median (min–max)
age	51.3 ± 13.8	55 (21–77)
sex (female)	16 (53)	
surgery	29 (97)	1–3 ops per patient
neurological deficit	11 (37)	
visual field cuts (amaurosis)	8 (27)	
hypopituitarism	19 (70)	
partial	15 (50)	
global	9 (20)	
dose (Gy)		
marginal (D _{min})	16.0 ± 2.1	16.0 (11.1–20.0)
max (D _{max})	28.9 ± 5.3	29.1 (16.4–40.0)
prescription isodose (%)	56.2 ± 7.6	55.0 (45.0–75.0)
TV (cm ³)	2.3 ± 2.2	1.7 (0.1–9.1)
isocenters per cm ³ TV	5.7 ± 5.3	5.1 (1–30)
prescription isodose volume/TV	1.26 ± 0.31	1.24 (0.80–2.25)

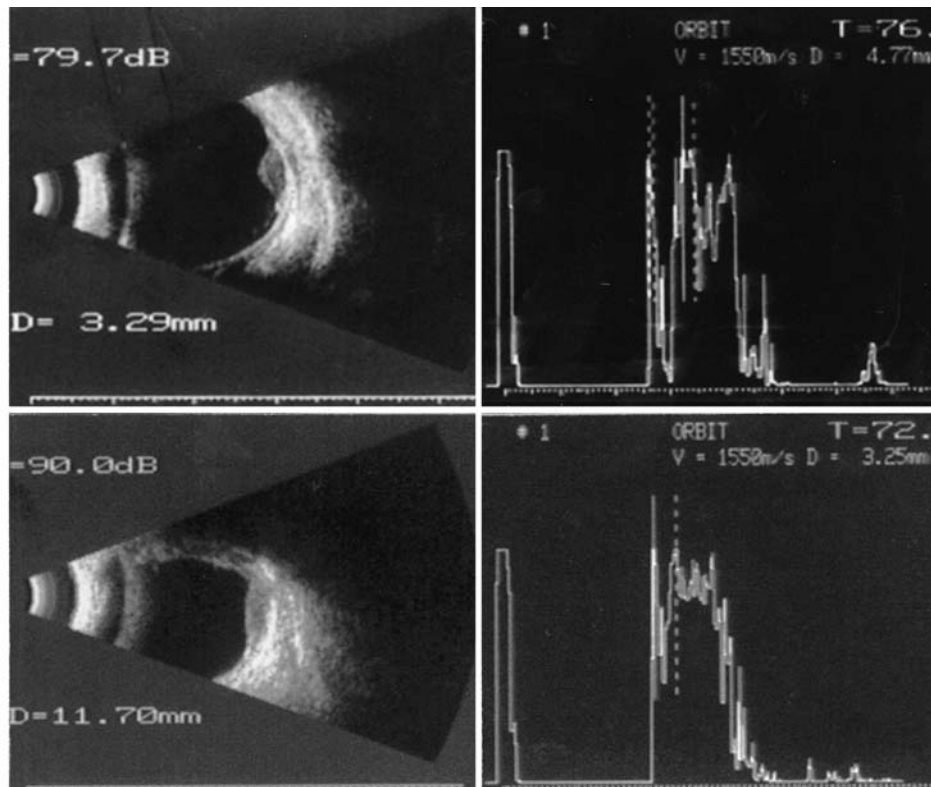


FIG. 2. Imaging studies obtained in the same case depicted in Fig. 1. Upper: The base of the tumor prior therapy was 10×9 mm (left) and its prominence was 4.8 mm (right). Lower: Image obtained 24 months after therapy (left), revealing a wider part of the tumor shown in Fig. 2 upper. Reduction of the tumor prominence to 3.25 mm with an increase of the inner reflectivity (right).

for small posteriorly located tumors and called this therapy transpupillary thermotherapy. A conventional diode laser with a specially designed adapter is used to raise the temperature of the tumor to 60°C . The diode laser penetrates the tumor to its base and can destroy it in one or more sessions. It is not clear how far the laser beam can penetrate, and according to Singh and Robertson (personal communication) an increasing incidence of recurrence will be reported in association with enucleation, because recurrent tumor grows into the orbit.

Endoresection Therapy

Damato, et al.,² and Peymann, et al.,¹⁰ have introduced endoresection for large or posteriorly located tumors not suitable for external resection. The rationale for this experimental therapy is that large tumors induced numerous complications, especially neovascular glaucoma. By using the vitrectomy probe, the tumor is aspirated and the detached retina is reattached in vitrectomy procedures. The disadvantages of this therapy are a possible spread of vital tumor cells, the risk of hemorrhage causing the loss of the eye, and a high number of retinal detachments.² The inventors of this new technique reported a recurrence in the sclera 1 year after its introduction.³ The recurrent tumor possibly arose from vital tumor cells in the sclera.

Bornfeld, et al.,¹ developed another procedure and combined single-fraction GKS with endoresection of the previously irradiated tumor. Prior to endoresection, a 20-Gy dose of radiation is administered in a single fraction at the

Krefeld institution and then the resection is performed at the Essen hospital. The authors speculated that the GKS may reduce the possibility of vital tumor cell spread.

Gamma Knife Radiosurgery

Mueller, et al.,⁷ have reported in their series of GKS-treated uveal melanomas, a tumor control rate of 98% after 3 years. Patients were treated with a margin dose of at least 25 Gy. Because these surgeons used a larger collimator with an inhomogeneous margin of healthy tissue, the 25-Gy margin dose is not comparable to ours. The advantage of using a large collimator is a relatively high isodose at the tumor margin. The disadvantage is that there is no defined margin dose because the isodoses are not confined to the tumor margin.

Our Experience

In our first series of 31 patients treated between 1992 and 1995 a high incidence of severe side effects was obvious. According to dose-volume calculations, the risk factors for severe side effects (especially neovascular glaucoma) were a large tumor volume, tumor proximity to the ciliary body, and a high dose. We have since steadily lowered the margin dose to 30 Gy in the 50% (60%) isodose in most patients. In most cases 8-mm collimators were used. In our opinion the better rate of tumor regression and globe preservation in the low-dose group is a matter of proper case selection. Tumors with a solid and diffuse por-

TABLE 2
Summary of clinical results stratified by group

Parameter	High Dose Group (%)	Low Dose Group (%)
globe preservation	22/31 (71)	24/33 (72)
tumor regression	26/31 (84)	28/33 (84)
complete regression (scar)	8/31 (26)	4/33 (12)
partial regression (50–80% in tumor height)	18/31 (58)	27/33 (81)
recurrence	5/31 (16)	5/33 (15)
neovascular glaucoma	15/31 (48)	3/33 (9)
proven metastasis	6/31 (19)	2/33 (6)

tion are excluded because there is an unacceptable chance that the diffuse part of the tumor will not be covered by the GKS dose. The reduction in severe side effects may also reflect better case selection. Cases involving ciliary body melanomas (with one exception) and tumors with more than a 10-mm tumor prominence and 16-mm base were excluded. Because we are treating cases are not suitable for brachytherapy with beta sources such as ruthenium 106, on the one hand, and many of the melanomas were located within 3 to 4 mm from radiosensitive structures such as the macula lutea or the optic nerve, on the other hand, we expect a high incidence of side effects in these patients.

It has been suggested (J Ganz, personal communication) that the optimal treatment plan could be achieved with 4-mm collimators. This would require, however, a larger number of shots and lead to an increased treatment time. Because the patient is often fixed in an uncomfortable prone position with extension of the head, this would render such treatment planning impractical in some of our patients.

Treatment Planning

In the planning procedure we define a gross volume corresponding to the macroscopically visible tumor as shown in the magnetic resonance images. The planning treatment volume is the gross volume combined with a safety margin of 1 to 2 mm. It is considered that a safety margin is important because we are dealing with a primary malignant tumor in which infiltration of the retina, invisible on magnetic resonance imaging, must be expected. Magnetic resonance imaging remains the imaging modality of choice because it alone can distinguish between tumor and serous retinal detachment.

Radiation Dose

The lowering of the dose has been associated with a marked decrease in severe side effects, especially neovascular glaucoma. The reason for recurrences is not clear. In both groups the incidence of recurrences was virtually identical, and thus it seems unlikely that the reduction of the dose is an important factor in relation to recurrence. Mueller, et al.,⁷ also reposted a success rate of 98% in their series of patients treated with the low dose of 25 Gy to the tumor margin.

Indications for Single-Fraction Radiosurgery

In our opinion macular melanomas and tumors of the midperiphery are excellent candidates for single-fraction

radiosurgery. Juxtapapillary melanomas, which are not suitable for transpupillary thermotherapy are associated with a high risk for recurrences because of surgical problems when suturing the radioactive plaque (either iodine-125 or ruthenium-106) into place. The same problem applies to macular or juxtamacular melanomas. Considering the risk for a higher incidence of recurrences on the one hand and radiation-related damage to the optic nerve on the other hand, we prefer the therapy with the highest chance of destroying the tumor in a single session. Even in cases of ruthenium-106 plaques with excellent results in terms of visual function, Lommatzsch, et al.,⁵ reported an increasing incidence of radiation-induced optic neuropathies after a longer follow up.

Conclusions

A reduction of the GKS margin dose to 30 Gy appears to reduce the incidence of serious side effects while not decreasing the tumor control rate. In our opinion a multicenter trial with a larger number of patients according to Good Clinical Practice could provide a definitive answer to the ideal dose much faster than a single-center pilot study.

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